

Ejercicios de Nmap (Redes y Seguridad Informática)

1. Comprobar conectividad básica y escaneo simple

nmap 192.168.0.1

```
(kali㉿kali)-[~]
$ nmap 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:19 CE
ST
Nmap scan report for 192.168.0.1
Host is up (0.0012s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    filtered  http
MAC Address: 00:A0:26:D2:68:9A (Teldat)

Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
```

- Escanea el host con IP 192.168.0.1 para detectar puertos abiertos comunes.

2. Escaneo de un rango de IP

nmap 192.168.0.1-10

- Escanea todas las direcciones IP del rango 192.168.0.1 a 192.168.0.10.

```
kali@kali: ~
Session Acciones Editar Vista Ayuda

(kali㉿kali)-[~]
$ nmap 192.168.0.1-10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 07:50 CE
ST
Nmap scan report for 192.168.0.1
Host is up (0.0020s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    filtered  http
MAC Address: 00:A0:26:D2:68:9A (Teldat)

Nmap scan report for 192.168.0.5
Host is up (0.00036s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open       ftp
23/tcp    open       telnet
80/tcp    open       http
139/tcp   open       netbios-ssn
514/tcp   open       shell
515/tcp   open       printer
631/tcp   open       ipp
7443/tcp  open       oracleas-https
8080/tcp  open       http-proxy
9100/tcp  open       jetdirect
MAC Address: 00:26:73:99:57:8C (Ricoh Company)

Nmap done: 10 IP addresses (2 hosts up) scanned in 29.76 seconds
```

3. Escaneo de varias direcciones IP específicas

nmap 192.168.0.1 192.168.0.5

```
(kali㉿kali)-[~]
└─$ nmap 192.168.0.1 192.168.0.5
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:25 CEST
Nmap scan report for 192.168.0.1
Host is up (0.00091s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    filtered  http
MAC Address: 00:A0:26:D2:68:9A (Teldat)

Nmap scan report for 192.168.0.5
Host is up (0.0012s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open       ftp
23/tcp    open       telnet
80/tcp    open       http
139/tcp   open       netbios-ssn
514/tcp   open       shell
515/tcp   open       printer
631/tcp   open       ipp
7443/tcp  open       oracleas-https
8080/tcp  open       http-proxy
9100/tcp  open       jetdirect
```

- Escanea múltiples IP individuales especificadas manualmente.

4. Escaneo de puertos específicos

nmap -p 1042 192.168.0.5

```
(kali㉿kali)-[~]
└─$ nmap -p 514 192.168.0.5
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:28 CE
ST
Nmap scan report for 192.168.0.5
Host is up (0.00050s latency).

PORT      STATE      SERVICE
514/tcp    open       shell
MAC Address: 00:26:73:99:57:8C (Ricoh Company)

Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds
```

- Escanea solo el puerto 1042 en la IP dada.
-

5. Escaneo de puertos UDP

`nmap -sU -p 7 11 15 18 19 20 21 22 51 143 514 8080 192.168.0.1`

```
(kali㉿kali)-[~]
└─$ nmap -sU -p 7 11 15 18 19 20 21 22 51 143 514 8080 192.168
.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:31 CE
ST
Nmap scan report for 192.168.0.1
Host is up (0.00060s latency).

PORT      STATE SERVICE
7/udp     closed echo
MAC Address: 00:A0:26:D2:68:9A (Teldat)

Nmap done: 12 IP addresses (1 host up) scanned in 10.82 second
s
```

- Escaneo en modo UDP de puertos específicos.
- El parámetro `-sU` indica que se trata de un escaneo de puertos UDP.

6. Escaneo del sistema operativo

`nmap -O -osscan-guess localhost`

```
(kali㉿kali)-[~]
└─$ nmap -O -osscan-guess localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:35 CE
ST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000058s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored
states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS deta
ils
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds
```

- Detecta el sistema operativo del host local.
- El parámetro `--osscan-guess` permite adivinar el sistema si no se reconoce con certeza.

7. Mostrar versión de los servicios

`nmap -sV -version-all localhost`

```
(kali㉿kali)-[~]
└─$ nmap -sV -version-all localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:36 CE
ST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored
states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect resul
ts at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.64 seconds
```

- Muestra la versión de los servicios detectados en el host local.

8. Mostrar información del sistema Nmap instalado

`nmap -SV --version-1ll localhost`

```
(kali㉿kali)-[~]
└─$ nmap -sV -version localhost
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.2 libssh2-1.11.1 libz-
1.3.1 libpcres2-10.45 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

- Posible errata tipográfica, probablemente se refiere a:

`nmap -sV --version-intensity 9 localhost`

9. Ver interfaces de red

`nmap --iflist localhost`

```
└─$ nmap --iflist localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:38 CE
ST
*****INTERFACES*****
DEV (SHORT) IP/MASK                TYPE      UP MTU  MAC
lo  (lo)    127.0.0.1/8             loopback  up 65536
lo  (lo)    ::1/128                loopback  up 65536
eth0 (eth0) 192.168.0.76/23          ethernet  up 1500 08:
00:27:6F:98:BB
eth0 (eth0) fe80::a00:27ff:fe6f:98bb/64 ethernet  up 1500 08:
00:27:6F:98:BB

*****ROUTES*****
DST/MASK                DEV  METRIC GATEWAY
192.168.0.0/23          eth0 100
0.0.0.0/0               eth0 100   192.168.0.1
::1/128                 lo   0
fe80::a00:27ff:fe6f:98bb/128 eth0 0
fe80::/64               eth0 1024
ff00::/8                eth0 256
```

- Muestra todas las interfaces de red y rutas de la máquina local.

10. Enviar paquetes TCP ACK

`nmap -sA 192.168.0.5`

```
(kali㉿kali)-[~]
└─$ nmap -sA 192.168.0.5
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 14:40 CE
ST
Nmap scan report for 192.168.0.5
Host is up (0.00077s latency).
All 1000 scanned ports on 192.168.0.5 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 00:26:73:99:57:8C (Ricoh Company)

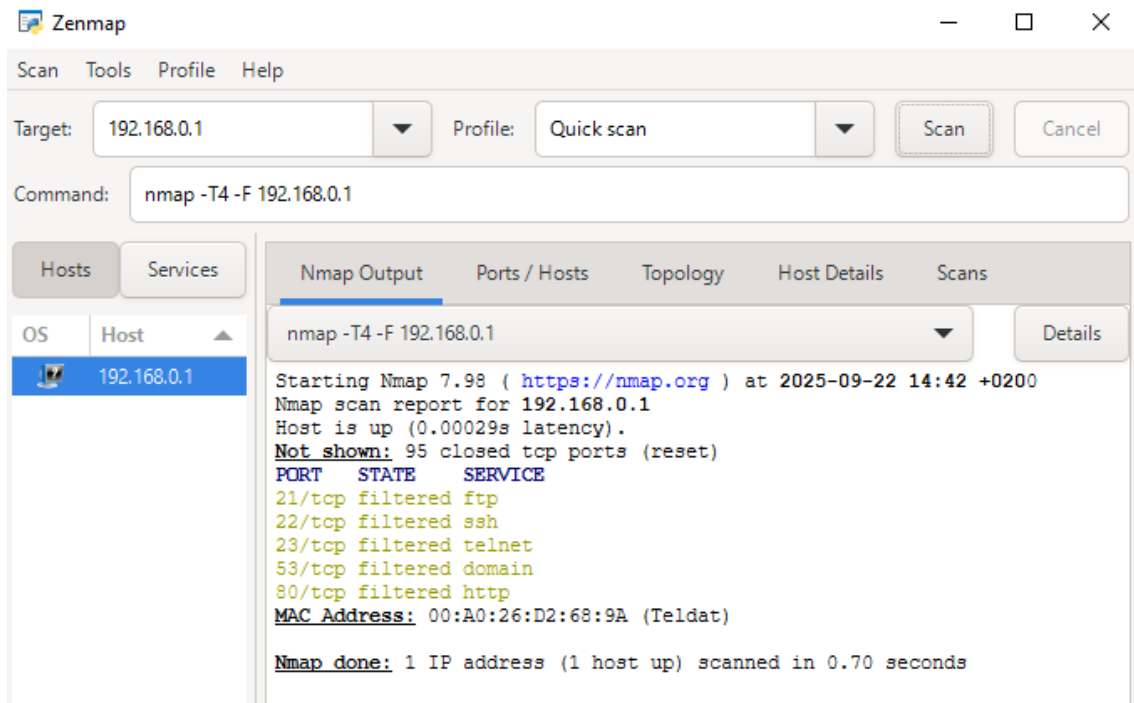
Nmap done: 1 IP address (1 host up) scanned in 23.92 seconds
```

- Escanea con paquetes TCP ACK para detectar hosts activos incluso tras firewall.
-

11. Actividad propuesta 1.9: Zenmap GUI para Nmap

🌿 Instrucciones del ejercicio:

Instala **Zenmap** (entorno gráfico para Nmap) en tu máquina Windows.

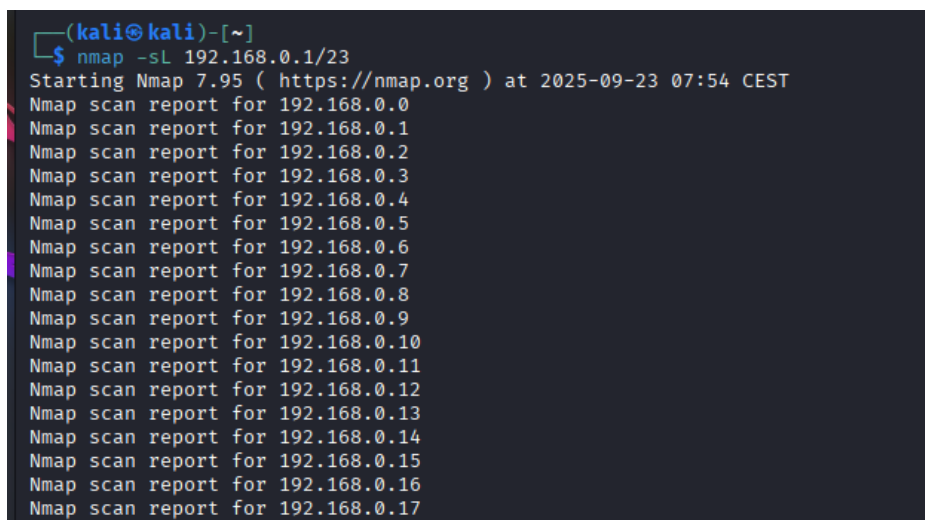


Investiga y explica **la utilidad de las diferentes pestañas** que ofrece la aplicación.

12. Otros comandos avanzados (mencionados en la imagen):

- Detección de DNS inverso:

`nmap -sL 192.168.0.1/23`



- Escaneo de versión con --version-trace

```
(kali㉿kali)-[~]  
$ nmap --version-trace  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 07:56 CEST  
PORTS: Using ports open on 0% or more average hosts (TCP:1000, UDP:0, SCTP:0)  
----- Timing report -----  
hostgroups: min 1, max 100000  
rtt-timeouts: init 1000, min 100, max 10000  
max-scan-delay: TCP 1000, UDP 1000, SCTP 1000  
parallelism: min 0, max 0  
max-retries: 10, host-timeout: 0  
min-rate: 0, max-rate: 0  
-----  
Read from /usr/share/nmap: nmap-protocols nmap-services.  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds
```

- Exclusión de host:

nmap 192.168.0.0/23 --exclude 192.168.0.31

```
(kali㉿kali)-[~]  
$ nmap 192.168.0.0/23 --exclude 192.168.0.31  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 07:58 CEST  
█
```